

# *Artificial Intelligence*

*An Advanced Course About Foundations*



## *First-Order Resolution*

Marco Piastra

# Propositional Resolution

a) Refutation  $\Gamma \cup \{\neg\varphi\}$  and translation into *conjunctive normal form* (CNF)

$\beta_1 \wedge \beta_2 \wedge \dots \wedge \beta_n$  where each  $\beta_i$  is a disjunction of literals (i.e.  $A$  or  $\neg A$ )

b) Translation of  $\Gamma \cup \{\neg\varphi\}$  in *clausal form* (CF)

$\{\beta_1, \beta_2, \dots, \beta_n\}$  where each  $\beta_i$  is a *clause* (i.e. a set of literals, representing a disjunction)

c) Exhaustive application of the resolution rule

1) Selection of two clauses  $\{\beta_1, \beta_2, \dots, \beta_n, \alpha\}, \{\neg\alpha, \gamma_1, \gamma_2, \dots, \gamma_m\}$

2) Generation of the *resolvent*

$\{\beta_1, \beta_2, \dots, \beta_n, \alpha\}, \{\neg\alpha, \gamma_1, \gamma_2, \dots, \gamma_m\} \vdash \{\beta_1, \beta_2, \dots, \beta_n, \gamma_1, \gamma_2, \dots, \gamma_m\}$

Termination conditions:

1) The empty clause has been derived (*success*)

2) No further resolutions are possible – *fixed point (failure)*

# Clausal Form in $L_{FO}$

1) Refutation:  $\Gamma \cup \{\neg\varphi\}$

2) Translation into *prenex normal form* (PNF):

All wff are now in the form:

$$Qx_1 Qx_2 \dots Qx_n \psi \quad (\text{the matrix } \psi \text{ does not contain quantifiers})$$

3) Removal of all existential quantifiers - *skolemization*:

All wff are now in the form:

$$\forall x_1 \forall x_2 \dots \forall x_m \chi \quad (\text{the skolemized matrix } \chi \text{ does not contain quantifiers})$$

Given that all wffs are universal sentences, the universal quantifiers can just be omitted

Example:

1:  $\forall x (P(x) \rightarrow (\exists y Q(x,y) \wedge R(y)))$

2:  $\forall x (\neg P(x) \vee (\exists y Q(x,y) \wedge R(y)))$

(removing  $\rightarrow$ )

2:  $\forall x \exists y (\neg P(x) \vee (Q(x,y) \wedge R(y)))$

(PNF)

3:  $\forall x (\neg P(x) \vee (Q(x, k(x)) \wedge R(k(x))))$

(Skolemization, with a new function  $k/1$ )

4:  $\neg P(x) \vee (Q(x, k(x)) \wedge R(k(x)))$

(omitting universal quantifiers)

Just atoms, connectives and parentheses...

# Clausal Form in $L_{FO}$

- 1) Refutation:  $\Gamma \cup \{\neg\varphi\}$
- 2) Translation into *prenex normal form* PNF :  
All wff are now in the form:  
$$Qx_1 Qx_2 \dots Qx_n \psi \quad (\text{the matrix } \psi \text{ does not contain quantifiers})$$
- 3) Removal of all existential quantifiers - *skolemization*:  
All wff are now in the form:  
$$\forall x_1 \forall x_2 \dots \forall x_m \chi \quad (\text{the skolemized matrix } \chi \text{ does not contain quantifiers})$$
  
Given that all wffs are universal sentences, the universal quantifiers can just be omitted
- 4) The *clausal form* can be obtained by just treating *atoms* as in *propositional logic*  
First translate in *conjunctive normal form* (CNF) and then in *clausal form* (CF)

Example:

- |                                                                  |                                |
|------------------------------------------------------------------|--------------------------------|
| 5: $\neg P(x) \vee (Q(x, k(x)) \wedge R(k(x)))$                  | (from before)                  |
| 6: $(\neg P(x) \vee Q(x, k(x))) \wedge (\neg P(x) \vee R(k(x)))$ | (CNF, by distributing $\vee$ ) |
| 7: $\{\neg P(x), Q(x, k(x))\}, \{\neg P(x), R(k(x))\}$           | (Clausal Form)                 |

# *Unificare necesse est, for resolution*

## ■ Problem: $\Gamma \models \varphi$ ?

$\Gamma \equiv \{\forall x (Greek(x) \rightarrow Human(x)), \forall x (Human(x) \rightarrow Mortal(x)), Greek(socrates)\}$

$\varphi \equiv Mortal(socrates)$

*Refutation, translation, clausal form:*

1:  $\{\forall x (Greek(x) \rightarrow Human(x)), \forall x (Human(x) \rightarrow Mortal(x)), Greek(socrates), \neg Mortal(socrates)\}$

(  $\Gamma \cup \{\neg\varphi\}$  is already in PNF, no skolemization is needed)

2:  $\{\{Human(x), \neg Greek(x)\}, \{Mortal(x), \neg Human(x)\}, \{Greek(socrates)\}, \{\neg Mortal(socrates)\}\}$

(Clausal Form)

*Resolution method (attempt):*

3: Try resolving:  $\{\neg Mortal(socrates)\}, \{Mortal(x), \neg Human(x)\}$

*Technically, no resolution is applicable: no pairs of complementary literals*

*Intuitively though,  
the two literals  $\neg Mortal(socrates)$  and  $Mortal(x)$  are complementary, somehow...*

# Unification

*Replacing variables with terms to render two atoms identical*

## ■ Unifier

A substitution of variables with terms  $\sigma = [x_1 = t_1, x_2 = t_2 \dots x_n = t_n]$  that makes two complementary literals  $\alpha$  and  $\neg\beta$  *resolvable*

That is, it makes the two atoms *identical*:  $\sigma(\alpha) = \sigma(\beta)$

- Obviously, a unifier does not necessarily exist:  
for instance,  $P(g(x, f(a)), a)$  and  $\neg P(g(b, f(w)), k(w))$  are not unifiable

## ■ MGU - *most general unifier*

It is the minimal *unifier* of  $\alpha$  and  $\neg\beta$

$$\text{MGU } \mu \Leftrightarrow \forall \sigma \exists \sigma' : \sigma = \mu \cdot \sigma'$$

Any other unifier can be obtained as a composition of  $\mu$

# Constructing the MGU

## ■ Martelli and Montanari's algorithm

Input:  $[s_1 = t_1, s_2 = t_2 \dots s_n = t_n]$  (a system of *symbolic* equations)

Procedure:

Exhaustive application of the following rules to the system of symbolic equations  
(each rule *transforms* the original system)

- |                                                                         |                                                                               |                                                                         |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| (1) $f(s_1, \dots, s_n) = f(t_1, \dots, t_n)$                           | <i>replace by the equations</i><br>$s_1 = t_1, \dots, s_n = t_n,$             |                                                                         |
| (2) $f(s_1, \dots, s_n) = g(t_1, \dots, t_m)$ where $f \neq g$          | <i>halt with failure,</i>                                                     | ← Applies even when<br>either $m$ or $n$ are 0<br>(i.e. with constants) |
| (3) $x = x$                                                             | <i>delete the equation,</i>                                                   |                                                                         |
| (4) $t = x$ where $t$ is not a variable                                 | <i>replace by the equation <math>x = t,</math></i>                            |                                                                         |
| (5) $x = t$ where $x$ does not occur in $t$<br>and $x$ occurs elsewhere | <i>apply the substitution <math>\{x/t\}</math><br/>to all other equations</i> |                                                                         |
| (6) $x = t$ where $x$ occurs in $t$ and $x$ differs from $t$            | <i>halt with failure.</i>                                                     |                                                                         |

Unless an explicit failure occurs (i.e. by rules (2) or (6)),  
the procedure terminates with success when no further rule is applicable

# Constructing the MGU: examples

Example:  $[f(x, a) = f(g(z), y), h(u) = h(d)]$

$[x = g(z), y = a, h(u) = h(d)]$

$[x = g(z), y = a, u = d]$

Rule (1) on  $f(x, a) = f(g(z), y)$

Rule (1) on  $h(u) = h(d)$ , MGU

Example:  $[f(x, a) = f(g(z), y), h(x, z) = h(u, d)]$

$[x = g(z), y = a, h(x, z) = h(u, d)]$

$[x = g(z), y = a, h(g(z), z) = h(u, d)]$

$[x = g(z), y = a, u = g(z), z = d]$

$[x = g(d), y = a, u = g(d), z = d]$

Rule (1) on  $f(x, a) = f(g(z), y)$

Rule (5) on  $x = g(z)$

Rule (1) on  $h(g(z), z) = h(u, d)$

Rule (5) on  $z = d$ , MGU

Example:  $[f(x, a) = f(g(z), y), h(x, z) = h(d, u)]$

$[x = g(z), y = a, h(x, z) = h(d, u)]$

$[x = g(z), y = a, h(g(z), z) = h(d, u)]$

$[x = g(z), y = a, g(z) = d, z = u]$

Rule (1) on  $f(x, a) = f(g(z), y)$

Rule (5) on  $x = g(z)$

Rule (2) on  $g(z) = d$  FAILURE



# Standardization of variables is also necessary

- Example:  $\Gamma \models \varphi$  ? (transitive property - in clausal form)

$$\{ \forall x \forall y \forall z (L(x,y) \wedge L(y,z)) \rightarrow L(x,z), L(a,b), L(b,c), L(c,d) \} \models L(a,d) \text{ ?}$$

$$\Gamma \equiv \{ \{ \neg L(x,y), \neg L(y,z), L(x,z) \}, \{ L(a,b) \}, \{ L(b,c) \}, \{ L(c,d) \} \}$$

$$\varphi \equiv \{ L(a,d) \}$$

*Refutation and resolution:*

- 1:  $\{ \{ \neg L(x,y), \neg L(y,z), L(x,z) \}, \{ L(a,b) \}, \{ L(b,c) \}, \{ L(c,d) \}, \{ \neg L(a,d) \} \}$
- 2: Unify and resolve  $\{ \neg L(x,y), \neg L(y,z), \textcolor{red}{L(x,z)} \}$  and  $\{ \neg \textcolor{red}{L(a,d)} \}$  :  
[ $x=a, z=d$ ] with resolvent  $\{ \neg L(a,y), \neg L(y,d) \}$
- 3: Unify and resolve  $\{ \neg L(x,y), \neg L(y,z), \textcolor{red}{L(x,z)} \}$  and  $\{ \neg \textcolor{red}{L(a,y)}, \neg L(y,d) \}$  :  
[ $x=a, z=y$ ] with resolvent  $\{ \neg L(a,y), \textcolor{blue}{\neg L(y,y)}, \neg L(y,d) \}$
- 4: *This seems to lead nowhere:  $\textcolor{blue}{\neg L(y,y)}$  will never be resolved in  $\Gamma \cup \{ \neg \varphi \}$*

*Why is this??*

# Standardization of variables is also necessary

- Example:  $\Gamma \models \varphi$  ? (transitive property - in clausal form)

$\Gamma \equiv \{ \{ \neg L(x,y), \neg L(y,z), L(x,z) \}, \{ L(a,b) \}, \{ L(b,c) \}, \{ L(c,d) \} \}$

$\varphi \equiv \{ L(a,d) \}$

*Refutation and resolution, standardize variables before each resolution*

(i.e. rename all variables with new, unique names)

- 1:  $\{ \{ \neg L(x,y), \neg L(y,z), L(x,z) \}, \{ L(a,b) \}, \{ L(b,c) \}, \{ L(c,d) \}, \{ \neg L(a,d) \} \}$
- 2: Unify and resolve  $\{ \neg L(x_1,y_1), \neg L(y_1,z_1), L(x_1,z_1) \}$  and  $\{ \neg L(a,d) \}$  :  
[ $x_1=a, z_1=d$ ] with resolvent  $\{ \neg L(a, y_1), \neg L(y_1,d) \}$
- 3: Unify and resolve  $\{ \neg L(x_2,y_2), \neg L(y_2,z_2), L(x_2,z_2) \}$  and  $\{ \neg L(a,y_3), \neg L(y_3,d) \}$  :  
[ $x_2=a, z_2=y_3$ ] with resolvent  $\{ \neg L(a, y_2), \neg L(y_2, y_3), \neg L(y_3,d) \}$
- 4: Unify and resolve  $\{ \neg L(a, y_4), \neg L(y_4, y_5), \neg L(y_5,d) \}$  and  $\{ L(a,b) \}$  :  
[ $y_4=b$ ] with resolvent  $\{ \neg L(b, y_5), \neg L(y_5,d) \}$
- 5: Unify and resolve  $\{ \neg L(b, y_5), \neg L(y_5,d) \}$  and  $\{ L(b,c) \}$  :  
[ $y_5=c$ ] with resolvent  $\{ \neg L(c,d) \}$
- 5: Resolve  $\{ \neg L(c,d) \}$  and  $\{ L(c,d) \}$  :  
resolvent  $\{ \}$   
(success)

# Resolution with unification for $L_{FO}$

A correct procedure for  $\Gamma \vdash \varphi$  in  $L_{FO}$

- a) Refutation  $\Gamma \cup \{\neg\varphi\}$ ,
- b) Prenex normal form and skolemization  $sko(\Gamma \cup \{\neg\varphi\})$
- c) Translation of  $sko(\Gamma \cup \{\neg\varphi\})$  into CNF hence into CF
- d) Repeat application of the resolution method:
  - 1) Selection of two clauses  $\{\beta_1, \beta_2, \dots, \beta_n, \alpha\}, \{\neg\alpha', \gamma_1, \gamma_2, \dots, \gamma_m\}$
  - 2) *Standardization* of variables  
(i.e. create new copies of the two clauses having new and unique variables)
  - 3) Construction of the MGU  $\mu$  (if it exists) for the two literals  $\alpha$  e  $\alpha'$
  - 4) Generation of the resolvent by applying of  $\mu$   
 $\{\beta_1, \beta_2, \dots, \beta_n, \alpha\}[\mu], \{\neg\alpha', \gamma_1, \gamma_2, \dots, \gamma_m\}[\mu] \vdash \{\beta_1, \beta_2, \dots, \beta_n, \gamma_1, \gamma_2, \dots, \gamma_m\}[\mu]$
- e) Until
  - 1) The empty clause has been derived (*success*)
  - 2) No further resolutions are possible – *fixed point (failure)*

Note: the method is not guaranteed to terminate (i.e. it might *diverge*)

# The method might diverge...

Problem:  $\{ \forall x (Q(f(x)) \rightarrow P(x)) \} \models \exists x (P(f(x)) \wedge \neg Q(f(x))) ?$

(The answer is negative: there is no entailment)

Refutation:

$\{ \forall x (Q(f(x)) \rightarrow P(x)) \} \cup \{ \neg \exists x (P(f(x)) \wedge \neg Q(f(x))) \}$

Prenex normal form:

$\{ \forall x (Q(f(x)) \rightarrow P(x)) \} \cup \{ \forall x \neg (P(f(x)) \wedge \neg Q(f(x))) \}$

(no skolemization required)

Clausal form:

$\{ Q(f(x)) \rightarrow P(x) \} \cup \{ \neg (P(f(x)) \wedge \neg Q(f(x))) \}$

$\{ \neg Q(f(x)) \vee P(x) \} \cup \{ \neg P(f(x)) \vee Q(f(x)) \}$

$\{ \{ \neg Q(f(x)), P(x) \}, \{ \neg P(f(x)), Q(f(x)) \} \}$

Resolution:

1:  $\{ \neg Q(f(x_1)), P(x_1) \}, \{ \neg P(f(x_2)), Q(f(x_2)) \}, [x_1/f(x_2)] \vdash \{ \neg Q(f(f(x_2))), Q(f(x_2)) \}$

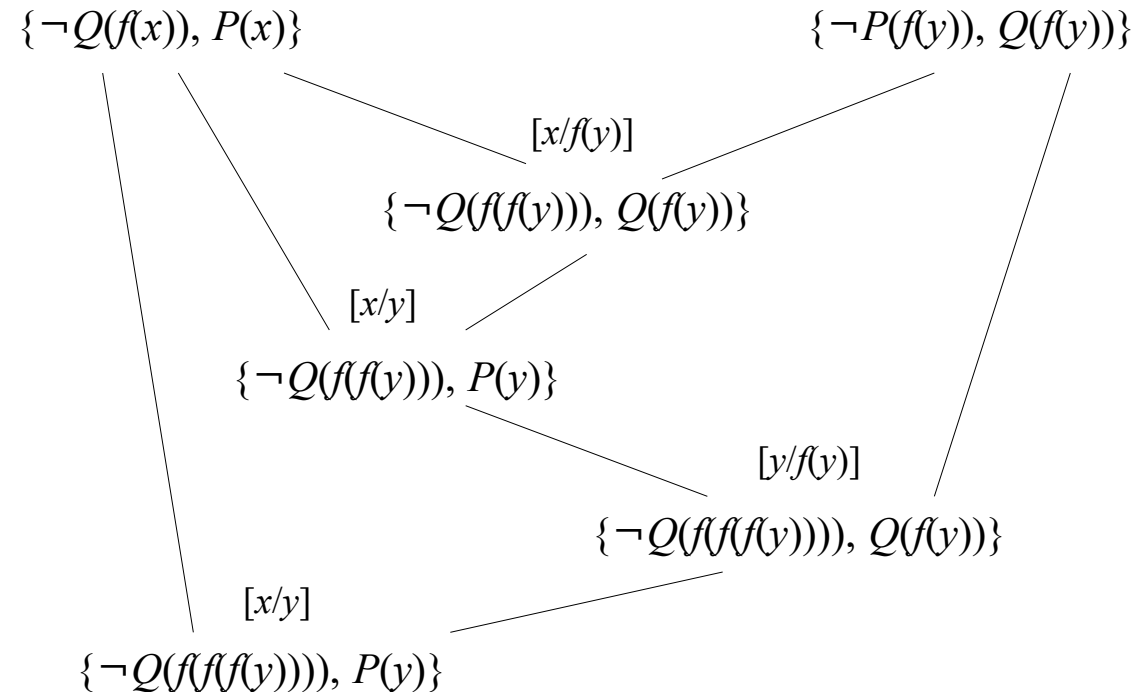
2:  $\{ \neg Q(f(x_3)), P(x_3) \}, \{ \neg Q(f(f(x_4))), Q(f(x_4)) \}, [x_3/x_4] \vdash \{ \neg Q(f(f(x_4))), P(x_4) \}$

3:  $\{ \neg Q(f(f(x_5))), P(x_5) \}, \{ \neg P(f(x_6)), Q(f(x_6)) \}, [x_5/f(x_6)] \vdash \{ \neg Q(f(f(f(x_6)))), Q(f(x_6)) \}$

4:  $\{ \neg Q(f(x_7)), P(x_7) \}, \{ \neg Q(f(f(f(x_8)))), Q(f(x_8)) \}, [x_7/x_8] \vdash \{ \neg Q(f(f(f(x_8)))), P(x_8) \}$

...

# The method might diverge...



· (Standardization of variables not shown, for simplicity)  
·  
·

# Properties of resolution with unification

- The method is *correct* in  $L_{FO}$

If the method finds the empty clause for  $sko(\Gamma \cup \{\neg\varphi\})$  then  $\Gamma \models \varphi$

- Is the method *complete* in  $L_{FO}$ ?

Within the limits of semi-decidability, yes (Robinson, 1963)

When  $\Gamma \models \varphi$ , the method will eventually find the empty clause for  $sko(\Gamma \cup \{\neg\varphi\})$

Very often (but not in the worst case) the method is more efficient than the one in the corollary of Herbrand's theorem

The advantage is due to *lifting*

(the method can resolve also non-ground clauses)

When  $\Gamma \not\models \varphi$ , the method might diverge

CAUTION: Unless the selection procedure is fair (more on this topic to follow) the method might diverge even when  $\Gamma \models \varphi$

Critical aspect:

- Selecting the clauses and literals to be resolved